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Simplified Procedure to Identify the Critical State Line of Crushable Rockfills

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Abstract: Particle breakage plays a relevant role on the stress-strain behavior of rockfills. As is widely recognized, the amount of particle breakage of coarse-grained soils depends on a combination of several factors, namely grain size, mineralogy, the shape of the particles, stress level, and degree of saturation and its variation. The results of monotonic triaxial tests carried out on a large apparatus on specimens of two types of rockfills and three different gradings highlighted the role of the coefficient of uniformity, grain tensile strength, and loading path on particle breakage and overall soil behavior. Starting from experimental evidence, a simplified conceptual model to predict the level of breakage during a generic stress path was developed. Experimental data supported the hypothesis of the influence of particle breakage on the position of the critical state lines (CSLs) in the compression plane. This implies that CSL parameters are state dependent, changing with current grading modifications, and are not intrinsic. A simplified procedure for predicting the position of the CSLs as a function of current particle breakage was, therefore, developed and calibrated against laboratory test results. **DOI: 10.1061/(ASCE)GT.1943-5606.0002668.** © *2021 American Society of Civil Engineers.*

Author keywords: Rockfill; Grain breakage; Triaxial testing; Critical state line (CSL).

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Introduction and Background

Rockfill materials are adopted in many civil engineering structures, such as embankments and dams. In order to ensure an appropriate performance of engineering structures, it is necessary to account for the main factors affecting the mechanical behavior of such coarse-grained soils. It is well known, for instance, that the large range of stresses involving rockfill structures can lead to relevant grain breakage that is the major cause of settlements or collapse.

From pioneering experimental research on large-scale apparatuses (Marsal 1967, 1973; Fumagalli 1969; Marachi et al. 1969, 1972), it has been recognized that grain crushing is mainly influenced by grain size, although other intrinsic characteristics (particle tensile strength, angularity, shape) and state parameters (induced stress level, stress path, relative density, moisture content) also have a significant influence (Lee and Farhoomand 1967; Hardin 1985; Hagerty et al. 1993; Lade et al. 1996; McDowell and Bolton 1998; Coop et al. 2004; Tarantino and Hyde 2005; Wood and Maeda 2008; Lirer et al. 2011; Casini and Viggiani 2013).

Various types of breakage indexes have been proposed in the literature to characterize the evolution of the grain size distribution caused by grain crushing (Marsal 1963; Hardin 1985; Lade et al. 1996; Einav 2007; Xiao and Liu 2017), but a recent study (Khonji et al. 2020) proved that Marsal's B_g factor is one of the most reliable. This parameter represents the percentage by weight of the solid subjected to breakage, and it is calculated by measuring the difference between the original and after crushing grading curves of the percentage retained for each sieve size. This difference is either positive or negative. Marsal's breakage factor B_g is the sum of the differences having the same sign (Lade et al. 1996).

Several authors (Hardin 1985; Hagerty 1993; Lade et al. 1996) have stated that, for a given crushable material, breakage factor increases with an increase in maximum particle size d_{max}, because bigger grains have a higher probability of being affected by defects acting as particle breakage triggers. Furthermore, it is well recognized (Lee et al. 1967; Lade et al. 1996) that grain breakage is linked to the coefficient of uniformity Uc $(= d_{60}/d_{10})$, which rules the number of interparticle contacts and, hence, the intensity of the forces exchanged by a single grain for a given stress level. In other words, a uniformly graded granular soil exhibits more crushing than a well-graded one (for a given value of d_{max}) because of the lower number of contacts and the larger average forces at the intergranular contact points. In addition, for the foregoing reason, grain crushing has a relevant effect on overall behavior, because it drives the original soil to more and more well-graded distributions, reducing the interparticle forces; as a consequence, during a generic loading path, soil becomes less prone to further crushing. The unavoidable and irreversible evolution of grading affects the mechanical response of rockfill in terms of shear strength, compressibility (Lirer et al. 2011), permeability, and critical state (Roscoe et al. 1958).

The strong link between particle breakage and soil grain has always represented a critical issue in the experimental study of rockfills; due to the limitations of traditional laboratory devices, the mechanical behavior of coarser materials is usually studied by modifying the original soil grading (by scalping or scaling techniques), choosing the maximum grain size d_{max} as a function of the specimen size D (Bishop and Henkel 1962; Penman 1971; Marsal 1973). From the foregoing discussion, it is clear that this choice has a relevant effect on the experimental study of the mechanical behavior of crushable materials.

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As previously mentioned, grading evolution has also a remarkable effect on the shape and location of the critical state line (CSL) in the compression plane $v - p'$ that, for an uncrushable soil, has the well-known analytical expression

$$\nu = \Gamma - \lambda \cdot \ln(p') \tag{1}$$

where ν = specific volume ($\nu = 1 + e$); p' = mean effective stress; and λ and Γ two parameters that control the slope and location of the CSL in the compression plane, respectively.

Although the nonlinearity of the CSL for granular soils at low stress level has been widely demonstrated (Been et al. 1991; Verdugo and Ishihara 1996; Li and Wang 1998; Carrera et al. 2011; Flora et al. 2012; Xiao et al. 2016), Eq. (1) is generally adopted for an ample range of stresses and refers to a given soil grading considered as an intrinsic soil property.

For granular soils subjected to grain breakage, the CSL has a more complex shape in the compression plane (Ghafghazi et al. 2014; Ciantia et al. 2019; Chang and Deng 2020), generally represented with a three-segment line for the full range of p' [Fig. 1(a)].

At very low stress levels, grain breakage is generally negligible, and soil behavior is controlled by changes in the particles arrangement. Above a breakage threshold (stress level $1 < p' < 3$ MPa), grain crushing becomes more pronounced and causes additional volume changes, resulting in a steeper slope of the CSL. At very high stress levels, when grain breakage stops, the CSL becomes flatter, and soil gradation does not change anymore. The three-segment line is identified by joining points at different stages of grading evolution; only the third line [the dashed line in Fig. 1(a)] is related to a stable grain size distribution. This phase is difficult to reach with traditional triaxial testing equipment, because grain breakage stops at very high deformation states (Luzzani and Coop 2002; Coop et al. 2004).

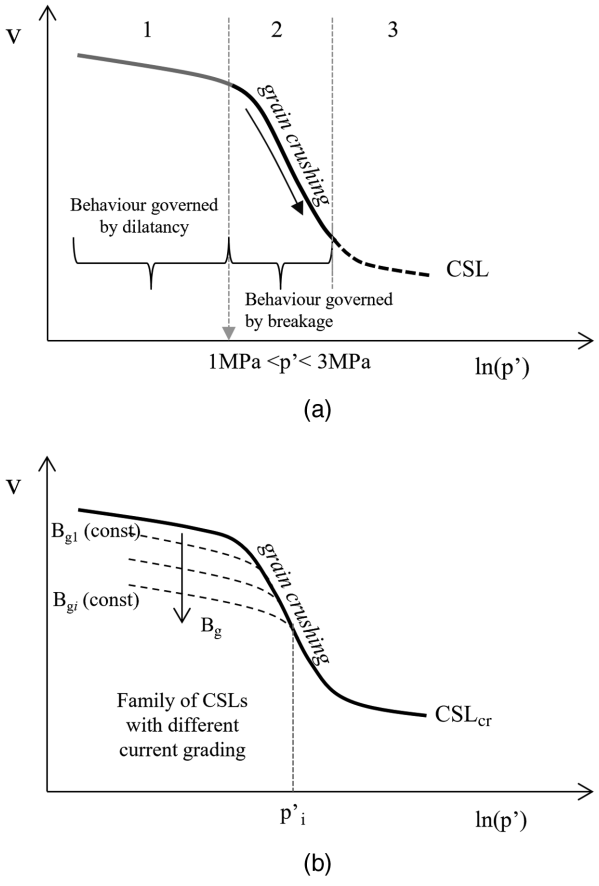


Fig. 1. (a) Three-segment CSL for crushable materials; and (b) family of CSLs changing with the current grading (B_g) in the compression plane. (Adapted from Chang and Deng 2020.)

All the aforementioned experimental evidence has led many authors (Coop and Lee 1993; Wood 2007; Wood and Maeda 2008; Xiao and Liu 2017; Chang and Deng 2020) to assume the existence of a multiplicity of CSLs in the compression plane correlated to the current grading [Fig. 1(b)]. Following this assumption, the concept of a critical state surface (defined in the v–p'–B_g space) has been introduced (Wood and Maeda 2008; Kikumoto et al. 2010; Xiao et al. 2016; Tengattini et al. 2016), explaining why the critical state line of crushable soils is not unique when represented by the projection of the aforementioned surface on the v–p' plane.

Also, the influence of particle breakage on the position of the CSL in the q–p' plane has been addressed by several authors. Bandini and Coop (2011) and Xiao et al. (2016) found that particle breakage results in the downward translation and rotation of the critical-state locus in the e–$\ln(p')$ plane without a change in the critical-state friction angle. On the contrary, Cheng et al. (2005), Liu et al. (2014), and Indraratna et al. (2015) observed that particle breakage decreases the critical stress ratio M. In addition, Yu (2017) suggested that the CSL is unique in the q–p' plane for original and precrushed sands in drained shear tests but moved downward with increasing particle breakage in undrained tests, implicitly indicating the coefficient of uniformity (and, therefore the number of contacts) as the responsible for the evolution of grading. It is evident that the nonuniqueness of the CSL due to the unknown (a priori) evolution of soil grading represents a relevant issue for all constitutive models that use the distance of the current volumetric state from the critical state [e.g., the state parameter ψ introduced by Been and Jefferies (1985)] to reproduce the mechanical behavior of granular soils. The experimental difficulties involved in attaining strain levels high enough to reach the critical state with traditional equipment (Lade et al. 1996; Russell and Khalili 2004) have led many authors to study the influence of particle breakage on critical state line location by means of discrete modeling approaches (Yan and Dong 2011; de Bono and McDowell 2014; Li et al. 2015; Tengattini et al. 2016; Xiao and Liu 2017; Ciantia et al. 2019). These studies have essentially confirmed the available experimental evidence (Daouadji et al. 2001; Russell and Khalili 2004; Wood and Maeda 2008; Xiao et al. 2015; Ciantia et al. 2019; Chang and Deng 2020), that is, that for lower stress levels ($p'_{max} < 3$ MPa), increasing particle breakage moves the CSL downward with negligible changes in λ [Fig. 1(b)]. The change in CSL slope ($\lambda \neq$ constant), however, becomes relevant at high stress levels in materials with low values of grain tensile strength (Bandini and Coop 2011; Chang and Deng 2020).

This is the reason why, in the stress range where breakage gradually becomes dominant over dilatancy, many authors have rearranged Eq. (1) for crushable soils as follows:

$$\nu = \Gamma(B_g) - \lambda \cdot \ln(p') \tag{2}$$

where the parameter Γ is a function of the level of grain crushing; and B_g = breakage index.

In this framework, the specific volume reduction due to grain breakage is superimposed on the volume change due to dilatancy (sliding and rolling of grains).

Although some advanced constitutive models have been developed (Daouadji et al. 2001; Russell and Khalili 2004; Kikumoto et al. 2010; Hu et al. 2011) to incorporate the effect of particle breakage on the stress-strain response of crushable soils, the point

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of uncertainty is how to predict the onset of breakage and the current intensity of particle crushing; the grading change is strictly connected to the applied stress history (deviatoric and/or isotropic components of the load), and, as a consequence, an evolution relation is needed to link the stress path to the resulting change in soil gradation (Hardin 1985; Russell and Wood 2009).

Starting from experimental evidence from monotonic drained tests performed on a large-scale triaxial apparatus on specimens of two rockfills with different grain size distributions, this paper introduces a simplified conceptual model to correlate grain crushing (represented by Marsal's particle breakage factor B_g) to the loading path. Within this framework, an analytical model to predict the progressively shifting CSLs in the compression plane is proposed—within a specific stress range—and verified against the experimental evidence from some crushable rockfills.

Laboratory Testing

Test Materials

The rockfills used in this study came from two different sites: a limestone quarry in Coreno Ausonio (Lazio region, Italy) and a heterogeneous rock mass close to the Fiumarella River (Calabria region, Italy).

The first soil (hereafter referred to as Coreno rockfill) was extracted by blasting and crushing a calcareous rock to the desired size. It is a low-grain-porosity, strong limestone with a specific gravity $G_{s,C} = 2.76$. It is typically employed as railway ballast or as a construction material for road and railway embankments. The single grains are irregular prisms with high roundness (i.e., its three dimensions are of the same order of magnitude), sharp edges, and high superficial roughness.

The second rockfill (hereafter referred to as Redisole rockfill) was used to build the Redisole Dam, a rockfill upstream-facing dam mainly built for irrigation purposes. The mineralogical composition of the original rock includes quartz, muscovite, calcite, and dolomite. The rockfill has an average specific gravity $G_{s,R} = 2.68$, with lightly elongated and planar grains (i.e., the three dimensions are of different orders of magnitude). This shape was expected to induce particle splitting on loading.

In order to study the effects of gradation and maximum particle diameter d_{max} on the location of the CSL, different grain size distributions were considered (Fig. 2). The values of d_{max} for each grading are 5 mm (fine grading curve), 30 mm (medium grading curve), and 30 mm (coarse grading curve), respectively. The fine and coarse gradings are two scaled grain size distributions so they have the same low value of U_c ($d_{60}/d_{10} = 1.85$). The medium curve ($U_c = 3.95$) refers to a well-graded rockfill having the same d_{min} as the fine curve and the same d_{max} as the coarse one.

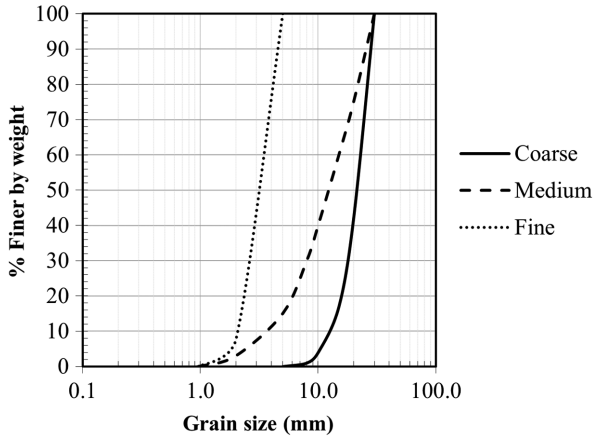


Fig. 2. Initial grain size distributions of the tested rockfills.

Table 1. Void ratios of Coreno and Redisole rockfill materials

Rockfill	Grain size distribution	D_{max} (mm)	D_{50} (mm)	U_c	e_{min}	e_{max}	e_o
Coreno	Coarse	30	21.40	1.85	0.606	0.904	0.695
	Medium	30	12.50	3.95	0.418	0.688	0.499
	Fine	5	3.38	1.85	0.610	0.826	0.675
Redisole	Coarse	30	21.40	1.85	0.659	0.862	0.720

All three gradings were tested for Coreno rockfill, while only the coarse grading was considered for Redisole rockfill (Table 1). For all gradings, the maximum and minimum void ratios were determined according to ASTM D4253-16e1 (ASTM 2016a) and ASTM D4254-16 (ASTM 2016b). As expected, the void ratios were quite similar for the parallel gradings (fine and coarse), highlighting that the soil fabric and the number of grain-to-grain contacts for a given soil are ruled by the coefficient of uniformity U_c.

For both rockfills, crushing tests were carried out at a constant displacement rate on groups of 3 grains of similar diameter d placed between two rigid and smooth steel plates (Fig. 3). The tensile strength of the grains p_f was calculated using the procedure suggested by Lee (1992); in the sequence of fractures that break the grains during loading (initially smoothening the contact points, then breaking larger and larger parts of the grains), p_f is conventionally defined as the tensile stress that causes the failure condition corresponding to the separation from the grain of fragment weighing more than 25% of the original grain. Given the load P measured at this stage, tensile strength p_f is defined as $P/(n \cdot d^2)$, where d = characteristic diameter of the group of grains, and n = average number of contacts between the particles and plates. Before starting the test, the average distance d between the two steel plates was calculated as the average of three measurements at 120°. This distance was assumed to be the characteristic diameter of the grains. Crushing tests were performed in wet and dry conditions on larger grains ($18 < d < 30$ mm) to check the relevance of moisture content (Ventini et al. 2020). The Coreno grains had a tensile strength p_f significantly higher than that of Redisole grains (Fig. 4). For both rockfills, p_f decreased significantly in wet conditions.

Testing Program

The tests were carried out on a large-scale triaxial apparatus designed and built at the University of Napoli Federico II (Ventini 2020). The device allows the testing of saturated and unsaturated specimens with a diameter D of 200 mm and a height H of 410 mm. Details on the drained triaxial tests performed on the Coreno and Redisole rockfill materials are given in Table 2. The first letter of the test ID refers to the material (C for Coreno and R for Redisole); the second letter refers to the grain size distribution (fine, medium, or coarse), and the number represents the confining stress σ_c' in 10 MPa. Some tests were carried out at constant σ_c', and some others were carried out at constant mean effective stress p', in which case the letter P was added to the ID. Initial confining stresses ranging from 0.05 to 0.70 MPa were investigated. Two isotropic compression (IC) tests were also carried out on a Coreno fine-grained specimen (labeled IC-CF in Table 2) and on a Redisole coarse-grained specimen (labeled IC-RC). In both cases, the saturated specimens were continuously loaded up 700 kPa and then unloaded to 100 kPa using a loading rate of 50 kPa/h.

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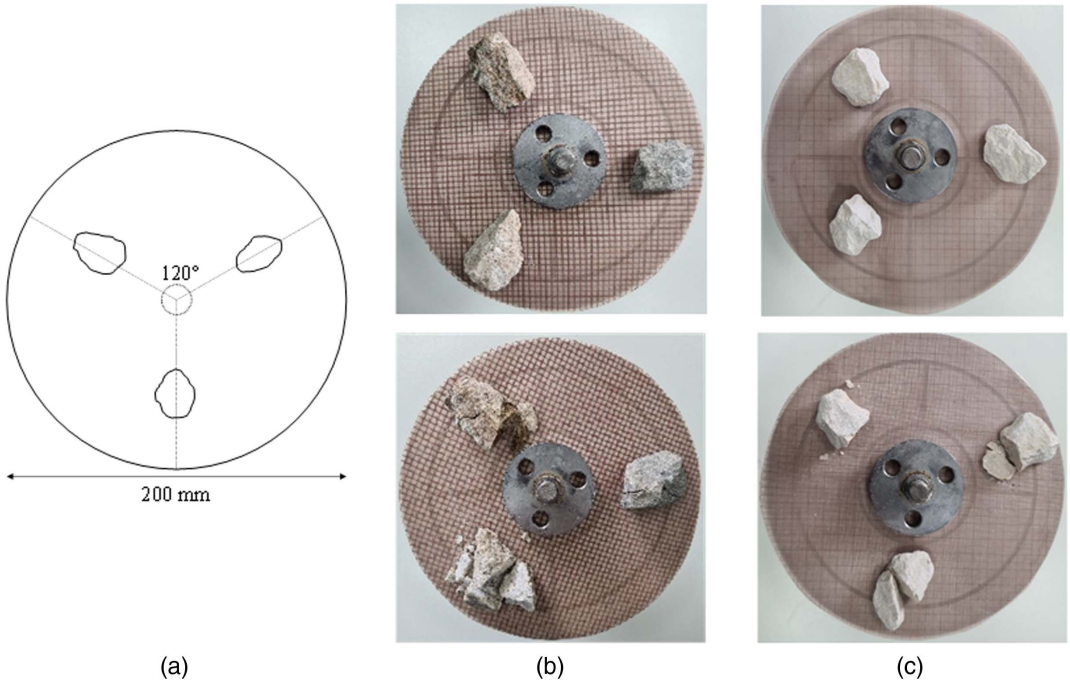


Fig. 3. Grain crushing tests: (a) arrangement of the grains; (b) Redisole grains before and after tests; and (c) Coreno grains before and after tests.

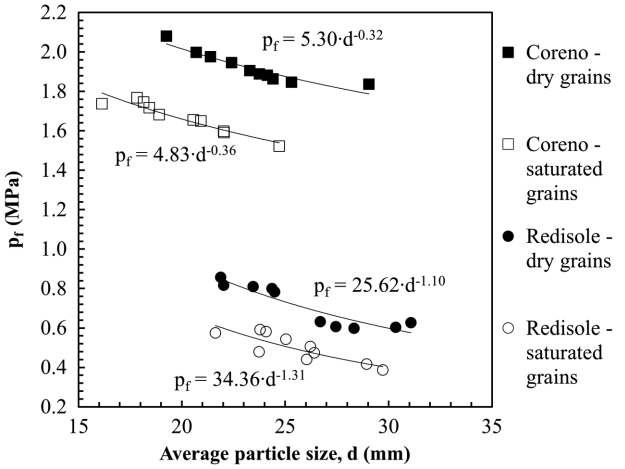


Fig. 4. Results of grain crushing tests on Coreno (C) and Redisole (R) rockfills. (Adapted from Ventini et al. 2020.)

In all tests, the specimens were compacted to a common $D_r = 70\%$ inside a split mold including a 3-mm-thick latex membrane. Compaction was performed by tamping five layers of the soil with a Marshall hammer. Particle breakage due to compaction was proven to be not relevant. In tests with $S_r = 1$, specimens were saturated by flushing, while in tests CF2U and CC2U, ambient humidity was used to consolidate the samples under $S_r = 0$. All the deviatoric stages of the tests were carried out at a constant axial displacement rate of 20.5 mm/h.

Stress Strain Behavior

The test results for the Coreno and Redisole rockfills are shown in Figs. 5–8 in the well-known planes: deviatoric stress q versus mean effective stress p'; deviatoric stress q versus axial strain ε_a; specific volume ν versus mean effective stress p'; and volumetric strain ε_v versus axial strain ε_a.

As expected, the rockfills showed a dilatant strain behavior affected by the adopted confining stress and the loading path. As is typical for coarse-grained soils (Flora 1995), the critical state condition was not reached in most tests, notwithstanding the high value of the final axial strain reached at the end of the tests ($\varepsilon_a > 20\%$).

Table 2. Details of consolidated drained triaxial tests on tested rockfills

Rockfill	Test ID	Grain size distribution	D_r	S_r	σ_c' (MPa)	Deviatoric phase p' (MPa)	Deviatoric phase σ_c' (MPa)
Coreno	IC-CF	Fine	0.7	1	up to 0.7	—	
	CF0.5				0.05	Variable	Constant
	CF2				0.2		
	CF4				0.4		
	CF5				0.5		
	CF2P				0.2	Constant	Variable
	CF6P				0.6		
	CF7P				0.7		
	CF2U			0	0.2	Variable	Constant
	CM0.5	Medium	0.7	1	0.05	Variable	Constant
	CM2				0.2		
	CM4				0.4		
	CM5				0.5		
	CM4P				0.4	Constant	Variable
	CC0.5	Coarse	0.7	1	0.05	Variable	Constant
	CC2				0.2		
	CC4				0.4		
	CC5				0.5		
	CC2U			0	0.2		
Redisole	IC-RC	Coarse	0.7	1	up to 0.7	—	
	RC0.5				0.05	Variable	Constant
	RC1				0.1		
	RC5				0.5		
	RC2P				0.2	Constant	Variable
	RC3P				0.3		

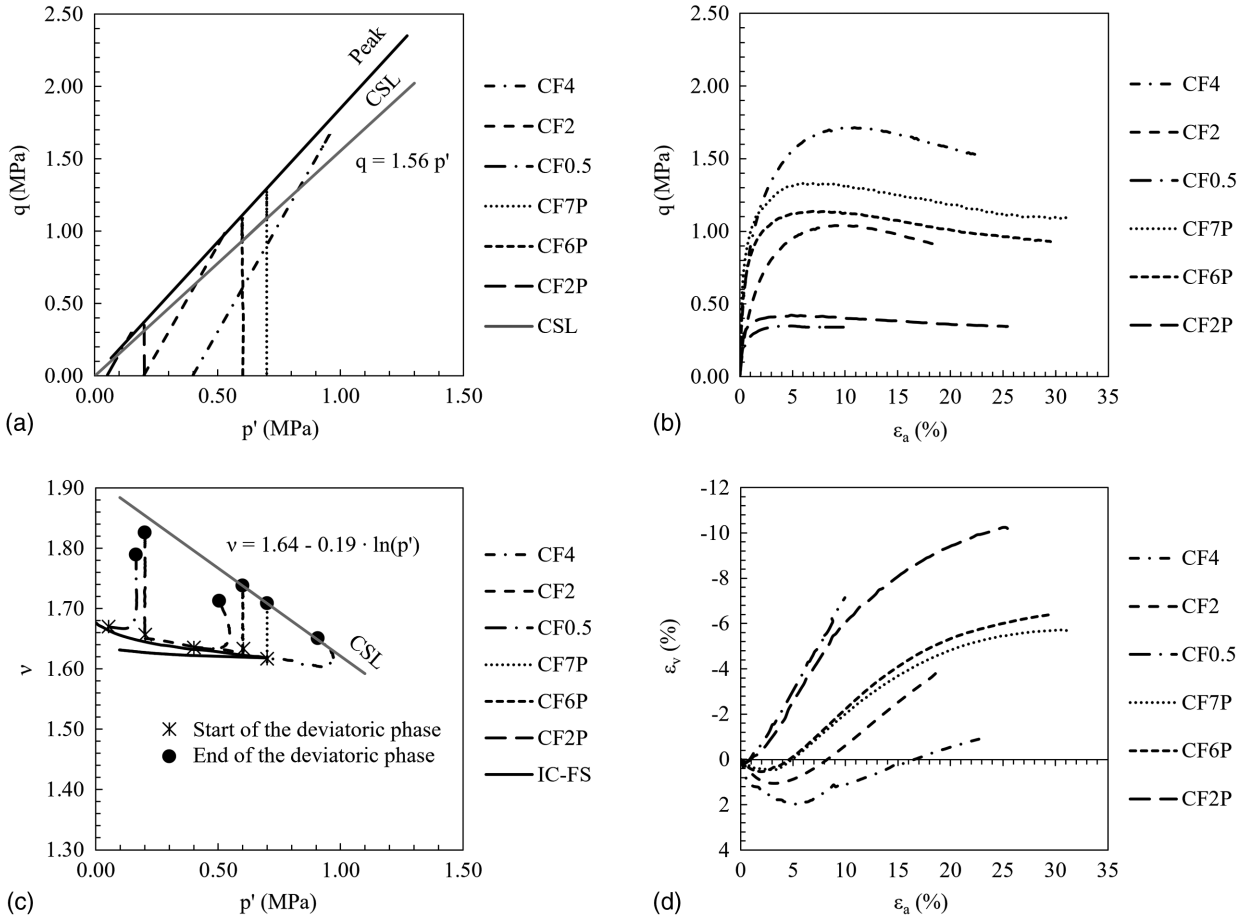


Fig. 5. Stress-strain behavior of Coreno rockfill—fine gradation.

For both the Coreno fine and medium graded specimens, the critical state was reached in at least two tests (namely CF6P and CF7P, and CM2 and CM4), allowing the identification of the CSL in the compression plane [Figs. 5(c) and 6(c)]. For the Coreno coarse graded specimens (Fig. 7), a lower strain level was reached due to the membrane punching that occurred in all the tests, probably due to particle breakage and, therefore, to the formation of roughness. For the Redisole rockfill, larger strain levels were reached in all tests and, as a consequence, critical state conditions were identified for three tests (i.e., RC1, RC2P, and RC3P; Fig. 8).

Particle Breakage Analysis

In order to quantify particle breakage, at the end of all the tests, the specimens were dried and subjected to a sieve analysis. To account for the effect of relative density on grain crushing, the dimensionless breakage parameter B_{gn} was chosen:

$$B_{gn} = B_g/(1 + e_{\text{cons}}) \tag{3}$$

where B_g = Marsal's breakage index; and e_{cons} = void ratio at the end of the isotropic compression phase of the tests.

Particle breakage during triaxial loading is due to both isotropic (spherical) and deviatoric stress components, the former mainly involving the smoothening of contact points and minor local breakage (at least at low to medium stress levels) and the latter causing major grain splitting (Fig. 9). Therefore, it can be expected that crushing by isotropic loading would mainly increase the fine content (only if the stress level is high enough compared to grain crushing strength; Fig. 4). On the contrary, deviatoric splitting causes more relevant effects on grading change.

The effect of the two components can be clearly identified in Fig. 10(a), which compares the Coreno grain size distributions measured at the end of the isotropic test for IC-CF (maximum confining stress $p' = 700$ kPa) and at the end of the triaxial drained test for CF7P (carried out at a constant mean stress $p' = 700$ kPa). The application of an isotropic load (IC-CF) caused a small variation in the original grading compared to the variation measured at the end of the deviatoric stage (CF7P).

In order to highlight the effect of grain crushing strength on breakage, the gradings after two tests performed on Coreno and Redisole coarse specimens following the same stress path are compared in Fig. 10(b). According to the results of the grain crushing tests (Fig. 4), the grading change is more evident for the Redisole specimen due to its tensile strength p_f being significantly lower than that of the Coreno rockfill.

As previously mentioned and verified in the grain crushing tests (Fig. 4), moisture condition also plays a role in the evolution of grading. As a matter of fact, the Coreno coarse specimen tested in wet condition [CC2; Fig. 10(c)] broke more than the specimen tested in dry condition [CC2U; Fig. 10(c)] along the same stress path. As recognized in the literature (Oldecop and Alonso 2001; Ventini et al. 2019), saturation increases grain crushing because of the weakening effect of water on grain-to-grain contacts, making crack propagation inside the grains easier and faster.

For Coreno rockfill, breakage was more pronounced for the fine and coarse graded specimens, while it was significantly lower for

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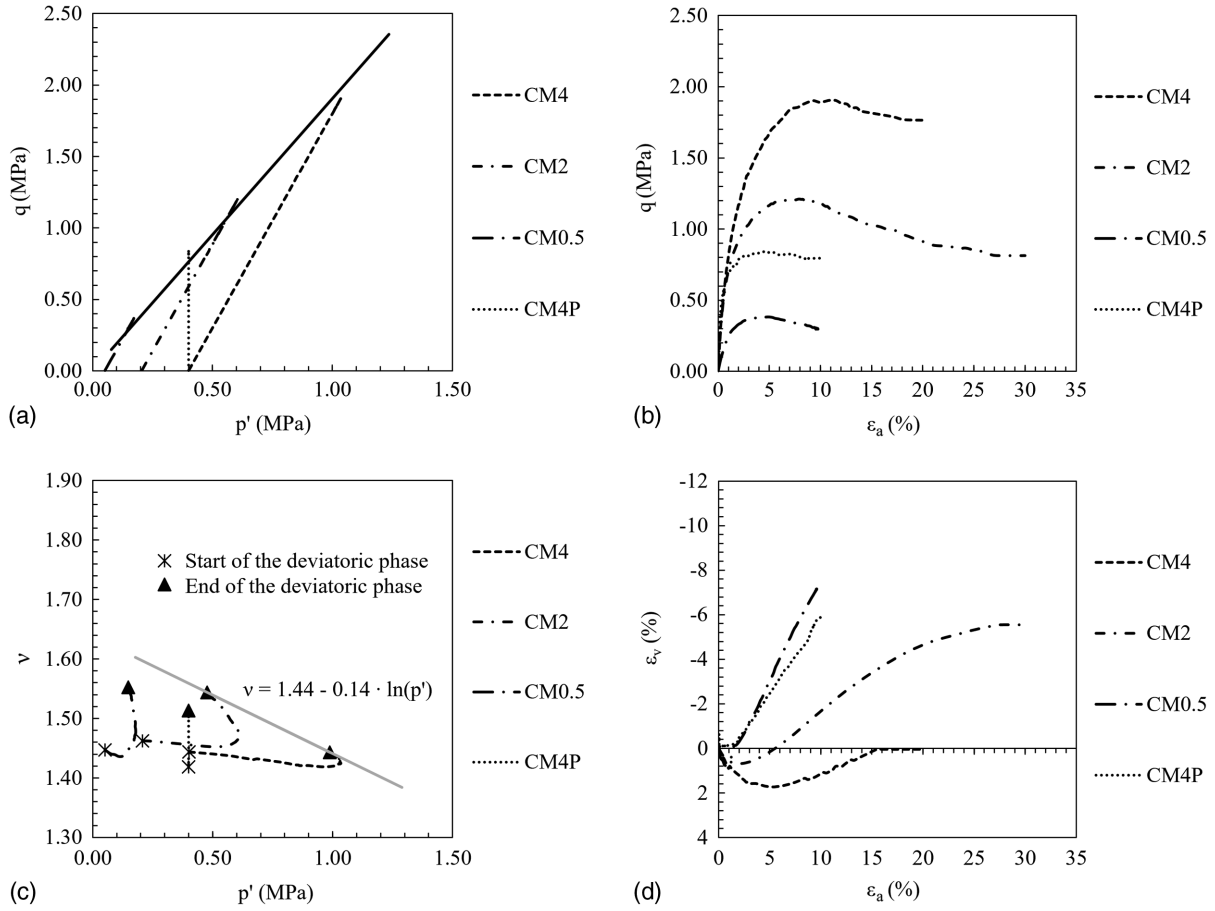


Fig. 6. Stress-strain behavior of Coreno rockfill—medium gradation.

the medium graded ones (Table 3). It is interesting that the medium graded rockfill suffered less crushing than the fine graded rockfill despite its larger soil particles. This can be explained by the concomitant effect of two factors: on the one hand, coarser particles crush more (Fig. 4), while, on the other hand, the better graded soil is characterized by a higher number of grain-to-grain contacts and lower average intergranular stresses. Therefore, for the Coreno rockfill, the effect of the coefficient of uniformity U_c on particle breakage was more relevant than the particle size effect. This is clearly shown in Fig. 11, in which grain size distributions before and after tests performed at a confining pressure of 0.4 MPa are reported for the three different gradings of Coreno rockfill.

Consistent with the foregoing qualitative explanation, the tests results can also be interpreted by comparing the average contact stresses acting on a single soil particle with grain crushing strength. To compute the average contact stresses $\sigma_{c,d}$ acting in the vertical direction on a single grain during the tests on Coreno rockfill, a simplified approach was used in this work (see Appendix). The results of triaxial tests on F, C, and M Coreno rockfills at $\sigma_c' = 0.05$, 0.20, and 0.40 MPa are plotted in Fig. 12 in terms of $\sigma_{c,d}$ versus B_{gn} relationship. For a given value of the macroscopic confining stress, the larger average contact stress acted on the coarse graded soil and the lower one acted on the medium graded rockfill. Consistent with the observed evidence for particle breakage, although the procedure adopted to quantify the contact stress was extremely simple, it was able to capture the influence of soil grading on particle breakage, returning $\sigma_{c,d}$ values always higher than the p_f evaluated using d_{50} of each grain size distribution (Fig. 2).

Simplified Model to Identify the CSL for Crushable Soils

General Framework

The experimental results highlighted that grain breakage is affected by both deviatoric and isotropic stress components. It is, therefore, reasonable to assume the existence of iso-B_g loci in the deviatoric plane q–p' [Fig. 13(a)]. A simple shape of such loci is an ellipse centered on the origin of the axes and having a shape described by the ratio between the major (a) and minor (b) axes. So, on the basis of the experimental data collected, only the right-hand part of the ellipse was considered in the model. The a:b ratio can be expected to change depending on the main effect of isotropic or deviatoric loading on particle crushing. The size of the ellipse (represented, for instance, by the actual a value) will grow as particle breakage increases, being, therefore, a state parameter; its shape (i.e., the a:b ratio) can be considered unchanging and representative of a (new) intrinsic material parameter. Such a parameter is certainly linked to single-grain properties (angularity, shape, mineralogy) as well as to the grain-to-grain contact stresses described, for example, by the uniformity coefficient U_c.

Using this conceptual framework, the same level of grain breakage B_{gn} can arise from a generic stress path ending on the same ellipse [e.g., AE, AF, AB in Fig. 13(a)]. The points belonging to the same ellipse in the q–p' plane [Fig. 13(a)] will have the same breakage factor (B_{g1}) and, therefore, must belong to the same critical state line CSL_{Bg1} in the ν–$\ln(p')$ plane [E'F'B'; Fig. 13(b)]. Another generic stress path AFGH reaches the initial ellipse

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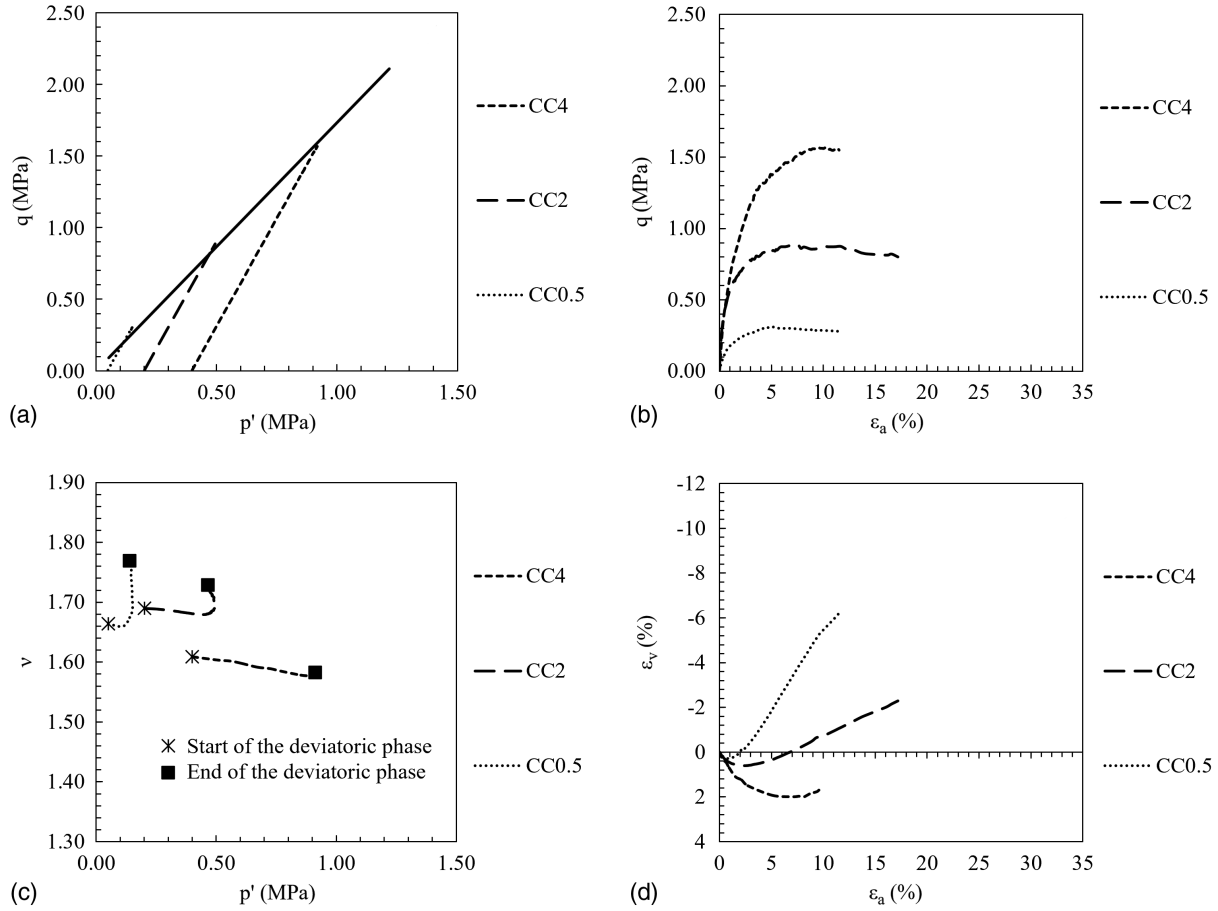


Fig. 7. Stress-strain behavior of Coreno rockfill—coarse gradation.

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($B_{g,1}$), moves it through different ellipses at increasing levels of breakage, and ends on the last ellipse ($B_{g,i}$) that has the same level of breakage reached with stress path ABCD. As a consequence, points H' and D' in the compression plane belong to the same line corresponding to the breakage index $B_{g,i}$.

In this framework, knowledge of the shape of the ellipse for a given soil (i.e, the a:b ratio) and the relation that links the growth of the ellipse with the level of particle crushing (i.e., B_g versus a), allows the identification of the CSLs at equal values of breakage index ($B_{g1}, B_{g2}, \ldots, B_{g,i}$) in the compression plane.

As previously stated [Fig. 1(b)], for the sake of simplicity it was assumed that particle breakage causes only a downward shift of the critical state line [λ = constant; Fig. 13(b)]. As a consequence, the gradient λ is regarded as an intrinsic parameter of the material. As stated by Wood and Maeda (2008), this is a consistent hypothesis for a wide range of stress levels.

Generally speaking, it can be considered that the location of the current CSL (B_{gn}) falls within two possible extreme CSLs expressed by the following equations [Fig. 13(b)]:

$$\nu = \Gamma_{\max} - \lambda \cdot \ln(p') \tag{4}$$

and

$$\nu = \Gamma_{\min} - \lambda \cdot \ln(p') \tag{5}$$

Therefore, the intercept $\Gamma(B_{gn})$ of the generic CSL varies between $\Gamma_{\min}$ and $\Gamma_{\max}$:

$$\Gamma_{\min} < \Gamma(B_{gn}) < \Gamma_{\max} \tag{6}$$

where the intercepts $\Gamma_{\min}$ and $\Gamma_{\max}$ may be linked to the minimum and maximum specific volumes of the initial grading (Table 1).

Saturated Tests

In this context, a possible analytical expression for the degradation law of the critical state line as a function of B_{gn} is as follows:

$$\Gamma(B_{gn}) = \Gamma_{\min} + (\Gamma_{\max} - \Gamma_{\min}) \cdot (1 + B_{gn}^{\alpha})^{-1} \tag{7}$$

where soil parameter α can be determined by means of a best-fitting procedure.

It is well known that void ratio range is affected by mean grain size, grain size distribution, and fine content. As a consequence, a change in void ratio range is expected for crushable soils. The model, however, was developed on the simplified hypothesis that the change in soil grading due to particle breakage does not affect the $v_{\max}$ and $v_{\min}$ values of the original soil grading (Table 1). This hypothesis can be considered reasonable for coarse soils ($1 < d_{50} < 20$ mm) in the stress range in which breakage gradually becomes dominant over dilatancy and the evolution of the initial grading is not extreme (Cubrinovski and Ishihara 2002; Chang and Deng 2020). Significant grain crushing has a greater effect on $v_{\min}$, but according to Cubrinoski and Ishihara (2002), at constant $v_{\min}$, $v_{\max}$ increases with the fine content. The proposed model starts from the identification of the shape of the ellipse, represented by the a:b ratio [Fig. 13(a)]. As stated previously, this aspect ratio

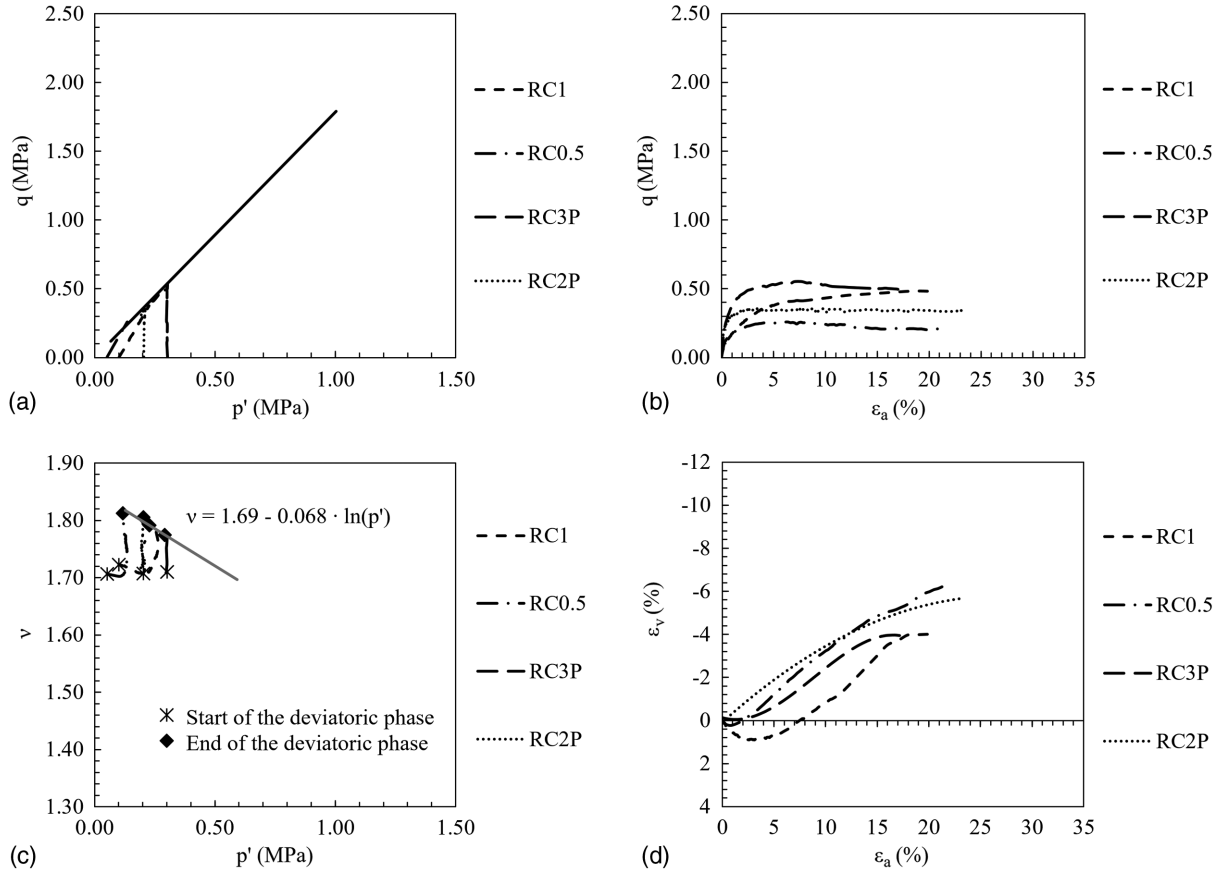


Fig. 8. Stress-strain behavior of Redisole rockfill—coarse gradation.

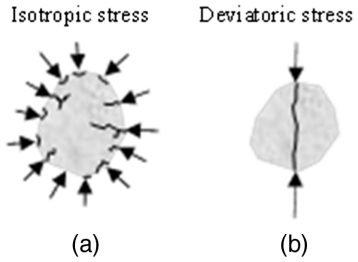


Fig. 9. Schematic of prevailing grain breakage mechanisms under different loading conditions: (a) diffused cracking and fine production under isotropic stress; and (b) grain splitting under deviatoric stress.

can be considered constant for the same material and must be determined by means of the equation of the ellipse passing through data points at equal B_{gn} in the q–p' plane [Fig. 13(a)].

In order to determine the aspect ratio of the Coreno rockfill, the results of three tests (IC-CF, CF0.5, and CF2P) (Table 2) were used, having similar breakage factor B_{gn} at the end of the tests (i.e., $B_{gn} = 1.41\%$, 1.60%, and 1.53%, respectively). The computed a:b ratio for the Coreno fine rockfill was 1.94 [Fig. 14(a)]. In order to verify this value, a further monotonic drained test was performed at a confining stress $\sigma_c' = 0.5$ MPa [CF5; Fig. 14(a)]. The test was stopped when the stress path reached the ellipse corresponding to B_{gn} equal to approximately 1.51%, and the grain size distribution was determined. The measured B_{gn} (1.56%) confirmed the theoretical

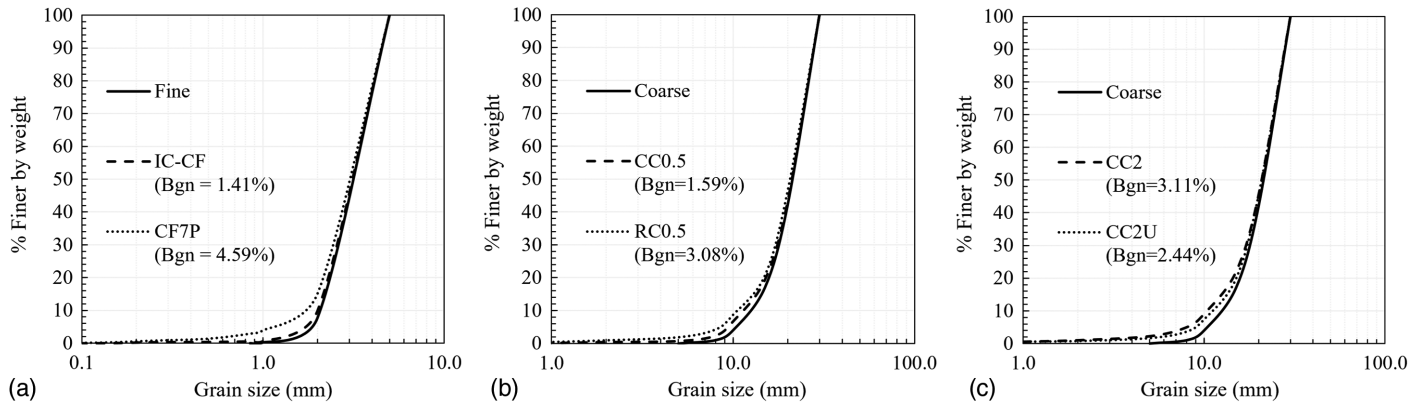


Fig. 10. Evolution of soil grading: effect of (a) stress path; (b) mineralogy; and (c) wetting path.

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Table 3. Particle breakage factors after performing monotonic tests

Rockfill	Test ID	Grain size distribution	S_r	σ_c' (MPa)	ν_{cons}	ν_{ult}	B_{gn} (%)
Coreno	IC-CF	Fine	1	up to 0.7	1.68	1.63	1.41
	CF0.5			0.05	1.67	1.79	1.60
	CF2			0.2	1.65	1.71	3.71
	CF4			0.4	1.64	1.65	6.03
	CF5			0.5	1.61	1.60	1.56
	CF2P			0.2	1.66	1.83	1.53
	CF6P			0.6	1.63	1.74	4.77
	CF7P			0.7	1.62	1.71	4.59
	CF2U		0	0.2	1.64	1.69	1.07
	CM0.5	Medium	1	0.05	1.45	1.55	1.60
	CM2			0.2	1.46	1.54	2.55
	CM4			0.4	1.44	1.44	2.98
	CM5			0.5	1.42	1.41	2.60
	CM4P			0.4	1.51	1.51	2.52
	CC0.5	Coarse	1	0.05	1.66	1.77	1.59
	CC2			0.2	1.62	1.73	3.11
	CC4			0.4	1.61	1.58	5.56
	CC5			0.5	1.61	1.61	3.14
	CC2U		0	0.2	1.62	1.69	2.44
Redisole	IC-RC	Coarse	1	up to 0.7	1.72	1.48	3.03
	RC0.5			0.05	1.71	1.81	3.08
	RC1			0.1	1.72	1.79	6.10
	RC5			0.5	1.71	1.70	3.06
	RC2P			0.2	1.71	1.81	3.47
	RC3P			0.3	1.71	1.77	6.20

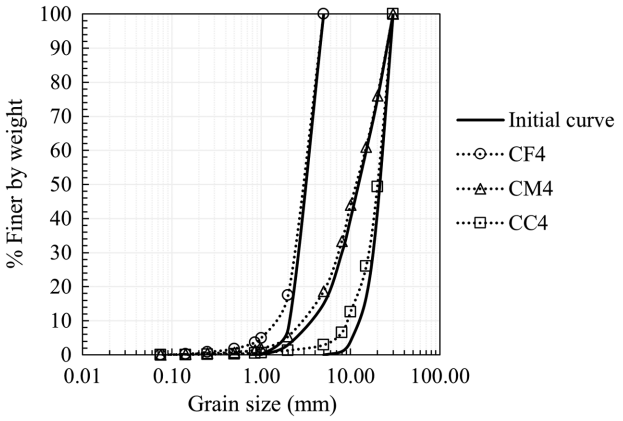


Fig. 11. Comparison of grain size distributions after monotonic tests at $\sigma_c' = 0.4$ MPa for Coreno rockfill.

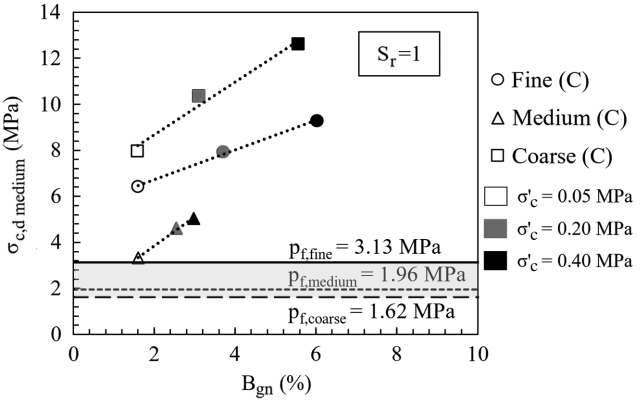


Fig. 12. Average vertical contact stress versus breakage factor in monotonic tests at $\sigma_c' = 0.05$, 0.20, and 0.40 MPa for Coreno rockfill.

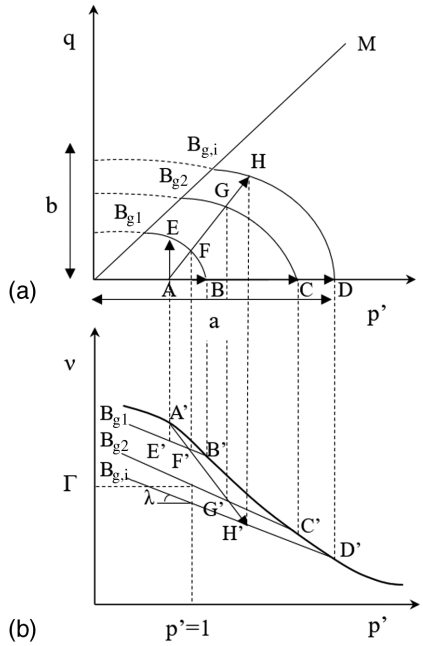


Fig. 13. Conceptual framework: surfaces at equal levels of grain breakage (a) in the deviatoric plane; and (b) in the compression plane.

value associated with the final ellipse achieved by the imposed stress path.

The same a:b aspect ratio identified for the Coreno fine rockfill was also used to interpret the tests performed on the medium and coarse gradings. Two other tests were carried out on the medium and coarse gradings [CM5 and CC5; Figs. 14(b and c)], confirming that the shape of the ellipse appears as an intrinsic soil property independent of grain size distribution (fine, medium, and coarse).

Following the same procedure, two tests (IC-RC and RC0.5, for which $B_{gn} = 3.03\%$ and 3.08%, respectively) were used to determine the aspect ratio of the Redisole rockfill (Fig. 15). For this material, the computed a:b ratio was 3.41, which was subsequently confirmed by the RC5 test ($B_{gn} = 3.06\%$). This result further shows that different soils (Coreno and Redisole rockfills) feature different aspect ratios, confirming the idea that a:b ratio is an intrinsic material property.

Interesting evidence arises from Fig. 16, in which the experimental data for B_{gn} pertaining to both Coreno rockfill (fine, medium, and coarse grading) and Redisole rockfill (coarse grading), were plotted against the value of the major ellipse axis a. A power regression can be used for both rockfills to link the growth of the ellipse with the breakage index B_{gn}. The parameters of the power law assume different values for the C and R materials, confirming that they are specific to the materials. As a confirmation of the foregoing statement, a single power regression applies to Coreno rockfill independent of the particular grading curves considered. Only one experimental point of the medium curve was not considered in the interpolation.

Once the shapes of the iso-B_g loci were determined for both tested rockfills, the next step consisted of identifying the CSLs (B_{gn}) in the compression plane [Eq. (2)]. As was previously stated,

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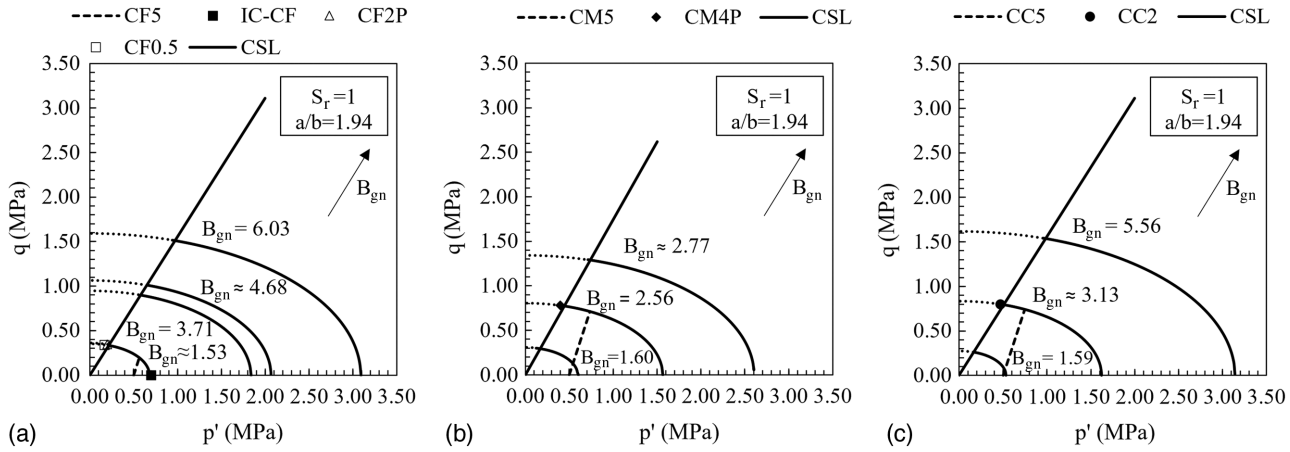


Fig. 14. Ellipses with $a{:}b = 1.94$ for triaxial tests on Coreno soil: (a) fine; (b) medium; and (c) coarse grain size distributions.

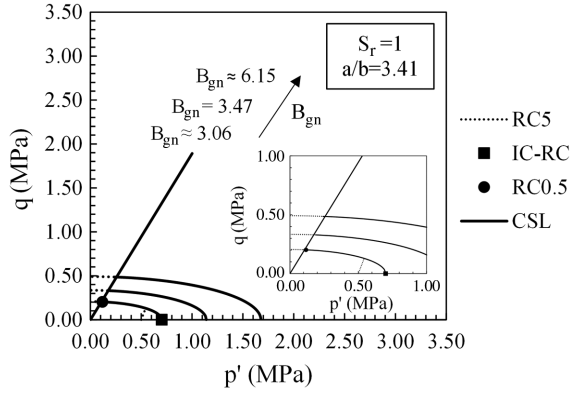


Fig. 15. Ellipses with $a{:}b = 3.41$ for triaxial tests on Redisole soil.

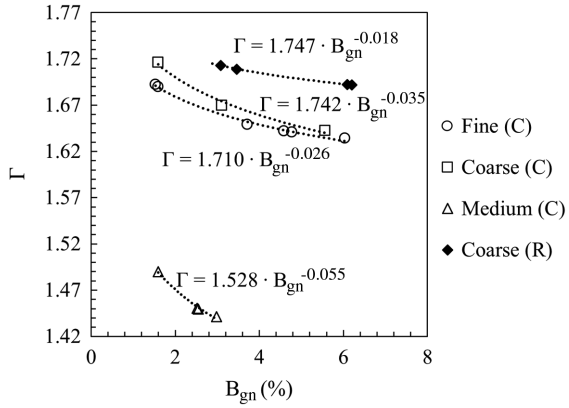


Fig. 17. Intercept Γ of the CSLs versus B_{gn} for saturated specimens of Coreno and Redisole rockfills.

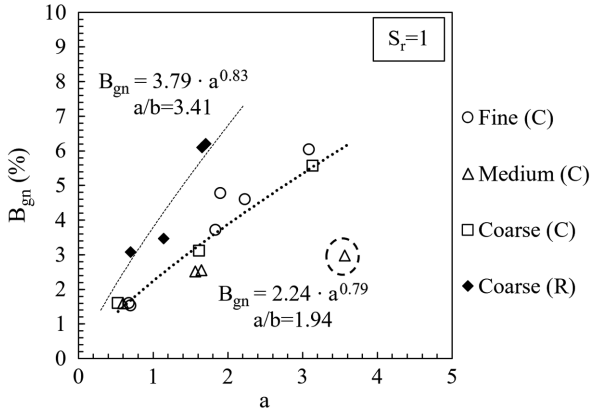


Fig. 16. Values of B_{gn} versus ellipse major axis a for the tested rockfill materials.

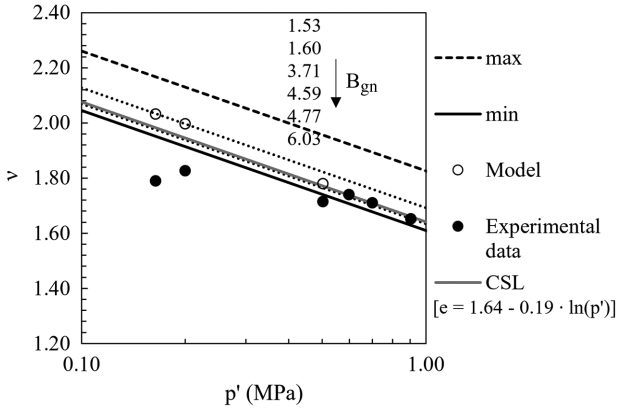


Fig. 18. CSLs for saturated specimens of Coreno fine rockfill.

the model hypothesis was that grain crushing does not change the slope λ of the CSL, which can be computed from tests that have reached the critical state condition.

As was previously shown, for Coreno fine grading, the position of the CSL [$\Gamma = 1.64$ and $\lambda = 0.19$; Fig. 5(c)] in the compression plane was identified by means of the CF6P and CF7P tests, which had the same level of grain crushing at the end of the tests (i.e., B_{gn} of approximately 4.68). Knowing Γ, B_{gn} and the limit intercepts Γ_{max} and Γ_{max} for these two tests, the value of α in Eq. (7) was determined by means of a best-fitting procedure ($\alpha = 1.15$).

Knowing the $\Gamma - B_{gn}$ relationship [Eq. (7); Fig. 17] and the gradient λ of the CSLs, allows the identification of the current location of the CSL(B_{gn}) in the compression plane and the prediction of the critical state void ratio e_{ss} even for tests not reaching this condition

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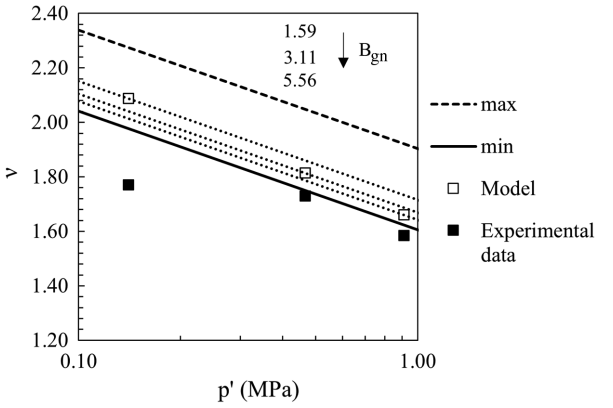


Fig. 19. CSLs for saturated specimens of Coreno coarse rockfill.

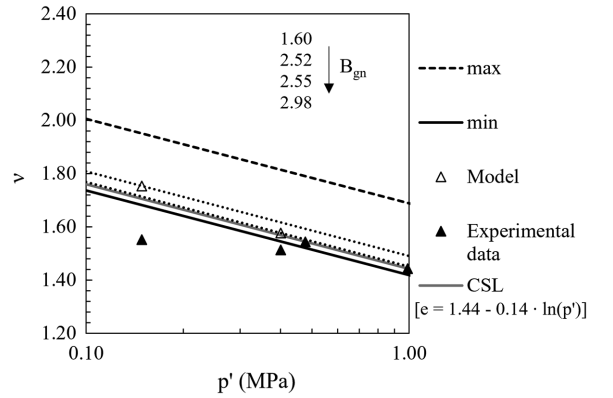


Fig. 20. CSLs for saturated specimens of Coreno medium rockfill.

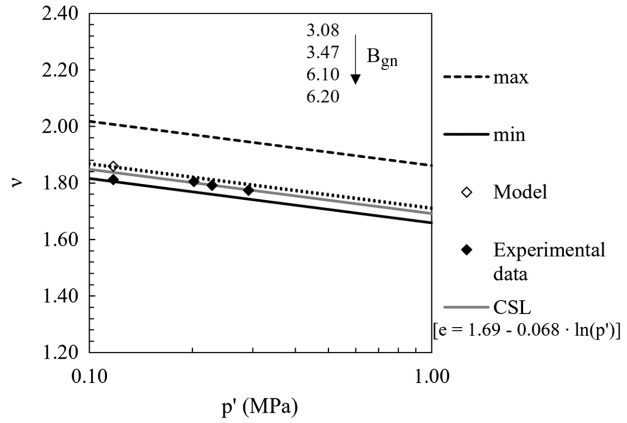


Fig. 21. CSLs for saturated specimens of Redisole coarse rockfill.

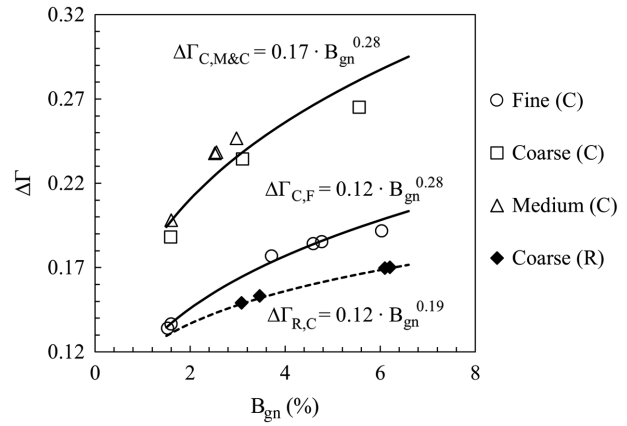


Fig. 22. Intercept range $\Delta\Gamma$ of the CSLs versus B_{gn} for saturated specimens of all tested rockfill materials.

(open circles in Fig. 18). As was observed in Figs. 5(b and d), two tests were very far from the critical state condition.

Given the unique $B_{gn} - \alpha$ relationship for the fine and coarse Coreno gradings (Fig. 16), the same model parameters identified for the fine grading ($\lambda = 0.19$ and $\alpha = 1.15$) were adopted to evaluate the $\Gamma - B_{gn}$ relationship (Fig. 17) and the CSLs for the coarse Coreno rockfill specimens (Fig. 19). Again, an experimental point was very far from the critical state (CS) condition. As for the fine grading, this was referred to the lowest confining stress test.

With reference to the medium grading, the position of the CSL [$\Gamma = 1.44$ and $\lambda = 0.14$; Fig. 6(c)] in the compression plane was identified by the CM2 and CM4 tests, which had a similar level of grain crushing at the end of the tests (i.e., B_{gn} approximately equal to 2.77). Knowing Γ, B_{gn}, and the limit intercepts Γ_{max} and Γ_{max} for these two tests, the value of α in Eq. (7) was determined by means of a best-fitting procedure ($\alpha = 2.16$). The results in terms of the Γ–B_{gn} relationship for the medium grading and the corresponding CSLs in the v– $\ln(p')$ plane are reported in Figs. 17 and 20, respectively. It is confirmed that the test at the lowest confining stress is the one farthest from the CS condition.

For the Redisole rockfill, the model was applied to extrapolate the final condition of the RC0.5 test (Fig. 8), the only test that ended before attaining the critical state condition. The model parameters were determined from the RC1 and RC3P tests ($\lambda = 0.068$ and $\alpha = 0.90$). The Γ–B_{gn} relationship (Eq. 7) and the corresponding CSLs are reported in Figs. 17 and 21, respectively.

Some further considerations arose from the analysis of the model predictions, shown in Fig. 22 in terms of $\Delta\Gamma$ ($\Delta\Gamma = \Gamma_{max}$–$\Gamma_{model}$) versus B_{gn} for all specimens of Coreno and Redisole rockfills.

The relationship between $\Delta\Gamma$ and B_{gn} can be generally expressed as

$$\Delta\Gamma = c \cdot B_{gn}^{d} \tag{8}$$

where c and d = two parameters that must be experimentally calibrated.

The results for Coreno rockfill (fine, medium, and coarse specimens) show that the parameter d ($d_C = 0.28$) is a constant of the material (not dependent on d_{max} and U_c), possibly related to particle tensile strength. This seems to be confirmed by the Redisole results ($d_R = 0.19$), for which the grain crushing tests (Fig. 4) revealed a tensile strength p_f significantly lower than that of the Coreno grains. Furthermore, the parameter c assumed the same value for the coarse and medium grading curves of Coreno rockfill and seems, therefore, to be linked to the maximum size of the grains (d_{max}). However, more experimental evidence is needed to confirm this outcome.

Unsaturated Tests

The model was adopted to predict the location of the CSL for tests carried out on unsaturated specimens, too (Table 2).

The stress-strain behavior exhibited by the unsaturated specimens of fine and coarse grading of Coreno rockfill is represented in Fig. 23 and compared with the results of the corresponding

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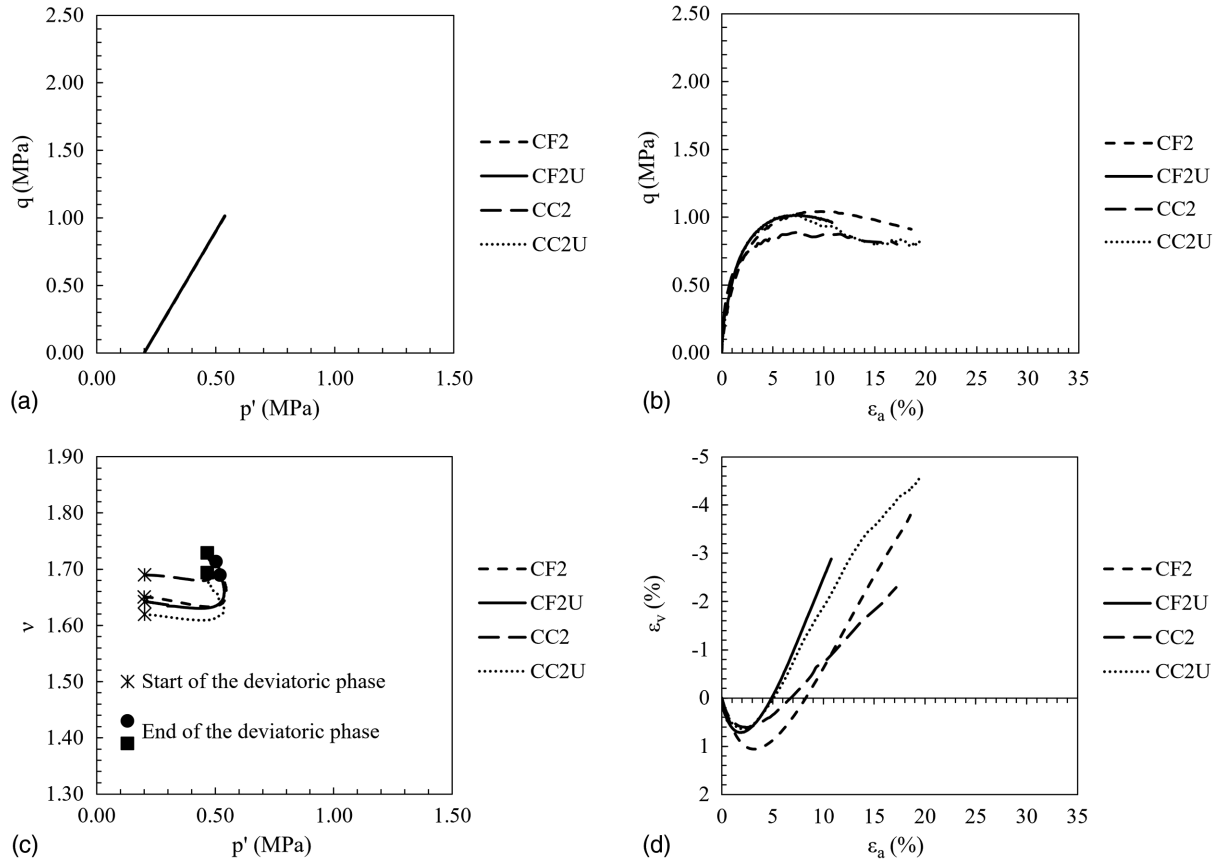


Fig. 23. Stress-strain behavior of Coreno rockfill—saturated versus unsaturated specimens.

saturated specimens. The presence of water did not imply considerable variation in the behavior in the q–p' plane [Fig. 23(a)] and in the q–ε_a plane [Fig. 23(b)]. All specimens showed a dilatant volumetric behavior [Figs. 23(c and d)], but none of the tests reached the critical state condition, despite the high strain levels attained at the end of the tests (ε_a approximately equal to 20%). For this reason, the location of the CSLs was identified by means of the proposed simplified model.

To this end, some simplified hypotheses were introduced. First, the shape of the ellipse (a:b ratio) and the parameters of Eq. (7) ($\Gamma_{\min}$, $\Gamma_{\max}$, and α) were considered independent of the degree of saturation; as a consequence, the same values identified for the saturated specimens were adopted for the corresponding unsaturated specimens. Furthermore, as stated by many studies (e.g., Wang et al. 2002; Kayadelen et al. 2007), in the compression plane the CSLs pertaining to unsaturated condition have the same slope of the ones pertaining to saturated condition (λ = constant). The results in terms of Γ–B_{gn} relationship for the fine and coarse unsaturated specimens (CF2U and CC2U) and the corresponding CSLs in the v–$\ln(p')$ plane are reported in Figs. 24 and 25, respectively. The data from unsaturated tests in the Γ–B_{gn} plane (Fig. 24) were in good agreement with the relationship determined for the saturated specimens, confirming the effectiveness of the proposed model. However, further experimental evidence is needed to confirm the results found within this study.

Model Validation on Independent Data

Next, a set of experimental literature results was used to check the proposed model. The data were taken from Varadarajan et al. (2003), who carried out triaxial tests on crushable coarse-grained materials from the Ranjit Sagar site. Three different grading curves derived from the original one by using the parallel gradation modeling technique (Lowe 1964) were considered by the authors. As a consequence, three different well-graded soils ($U_c = 120$) were selected, choosing three values of the maximum particle size ($d_{\max} = 25$, 50, and 80 mm). Basic information on the consolidated drained triaxial tests performed and the corresponding B_{gn} values are reported in Table 4.

As a first step, the hypothesis of an elliptical iso-B_g locus was verified. Because there were not enough available data to calculate the a:b ratio, a value of a:$b = 1.94$, close to the value found for the Coreno rockfill, was adopted. As was verified for the Coreno data, it was reasonable to assume that the a:b ratio had the same value for all three grain size distributions of the Ranjit Sagar rockfill.

Once the shape of the ellipse was fixed, the relationship linking its increase in size (represented by a) to grain breakage factor B_{gn} was plotted in Fig. 26. As was previously shown in Fig. 16, a single power regression fits all the results pertaining to Ranjit Sagar rockfill, confirming again that the B_{gn}–a relationship is intrinsic to a material and is not linked with the adopted grain size distribution. This outcome is obviously independent of the value of a:b used to derive the results plotted in Fig. 26.

The comparison between the Coreno and Ranjit Sagar data revealed that, as expected, the B_{gn}–a correlation is mostly governed by grain tensile strength.

Regarding the slope of the CSL in the compression plane, values of λ for the Ranjit Sagar rockfill ($0.005 < \lambda < 0.026$) were much lower than those of the Coreno rockfill ($0.14 < \lambda < 0.19$). This is consistent with the fact that poorly graded soils (such as the Coreno rockfill) tend to have steeper CSLs than well-graded ones (Been and Jefferies 1985; Ciantia et al. 2019). Knowing the

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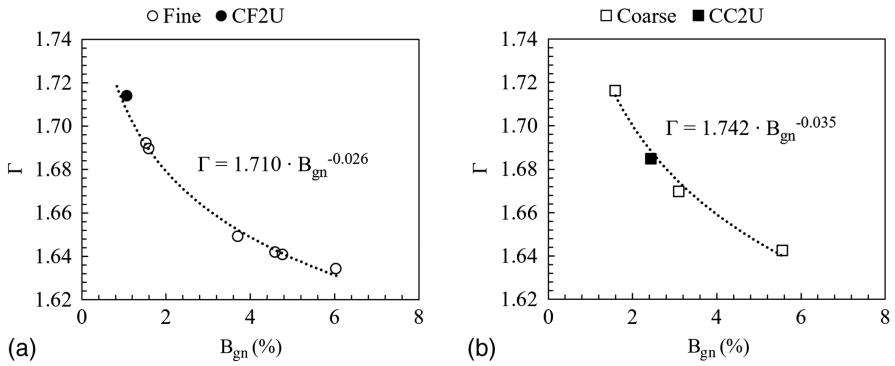


Fig. 24. Intercept Γ of the CSLs versus B_{gn} for saturated and unsaturated specimens of (a) fine; and (b) coarse Coreno rockfill.

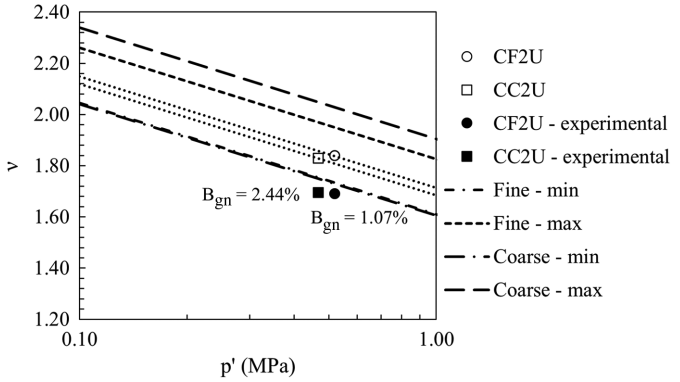


Fig. 25. Critical state lines determined with the proposed model for unsaturated specimens of Coreno rockfill.

gradient of the CSLs pertaining to the Ranjit Sagar rockfill, the downward shift of the CSL provided by the model [Eq. (7)] is represented in Fig. 27 in terms of values of $\Delta\Gamma$ ($\Delta\Gamma = \Gamma_{max} - \Gamma_{model}$) versus B_{gn}. The application of the proposed model to the literature data confirmed that the shift of the critical state line in the compression plane (represented by the $\Delta\Gamma$ parameter) is affected by the initial soil grain size distribution. As was observed for the Coreno rockfill (Fig. 22), the parameter c (Eq. 8) seems to depend on d_{max}. In contrast, the constant d of Eq. (8) is not related to grading, although both d_{max} and U_c of the Ranjit Sagar rockfill are very different from and, in particular, higher than those for the Coreno rockfill. Therefore, d is an intrinsic material property, as observed in the wide range of d_{max} and U_c in which the analysed two rockfills fall.

The reliability of the proposed model for the identification of the current location of the CSL in the compression plane can be verified in Fig. 28, in which the peak shear friction angles φ'_{peak} for drained simulations under both constant σ'_c and constant p' conditions were plotted versus state parameter ψ (Been and Jefferies 1985), defined as the difference between the void ratio at the end of the consolidation phase e_{cons} and the void ratio on the current critical state line at the same mean stress, computed by means of the simplified model (Figs. 18–20). The experimental data from triaxial compression tests reported by Varadarajan et al. (2003) were also included in Fig. 28. It is clear that the general trend in both cases, regardless of the grain size distribution, is a decrease in φ'_{peak} with increasing ψ (Been and Jefferies 1985; Huang et al. 2014; Ciantia and O'Sullivan 2020). The Coreno data had a general trend that was less inclined in the φ'_{peak}–ψ plane than the Ranjit Sagar data, probably due to the significantly different U_c values.

Concluding Remarks

Grading evolution due to particle breakage caused by loading and its consequences on the location of the CSL in the compression plane were investigated for two different rockfills via some monotonic drained triaxial tests carried out on a large apparatus.

The experimental data allowed the calibration of a simple conceptual model capable of predicting the evolution of particle breakage along generic stress paths and the corresponding shift in the location of the initial CSL ($B_g = 0$) in the compression plane v–$\ln(p')$.

Table 4. Details of triaxial tests on Ranjit Sagar rockfill

Rockfill	U_c	G_s	e_{max}	e_{min}	ν_{cons}	σ'_c (MPa)	p'_{ult} (MPa)	q_{ult} (MPa)	ν_{ult}	B_{gn} (%)
Ranjit Sagar (25 mm)	120.00	2.68	0.37	0.20	1.22	0.35	0.80	1.35	1.20	3.29
						0.70	1.47	2.30	1.20	4.11
						1.10	2.25	3.45	1.19	6.08
						1.40	2.87	4.40	1.19	7.15
Ranjit Sagar (50 mm)			0.38	0.20	1.22	0.35	0.93	1.75	1.20	4.11
						0.70	1.78	3.25	1.19	5.42
						1.10	2.77	5.00	1.19	7.39
						1.40	3.53	6.40	1.18	8.13
Ranjit Sagar (80 mm)			0.35	0.15	1.17	0.35	1.12	2.30	1.15	5.11
						0.70	2.10	4.20	1.15	6.90
						1.10	3.23	6.40	1.15	8.35

Source: Data from Varadarajan et al. (2003).

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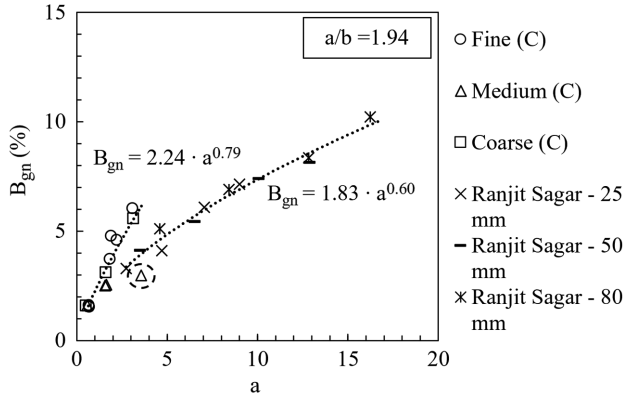


Fig. 26. Values of B_{gn} versus size of the grain breakage locus (ellipse major axis a) for the tested Coreno rockfills and for Ranjit Sagar materials. (Data from Varadarajan et al. 2003.)

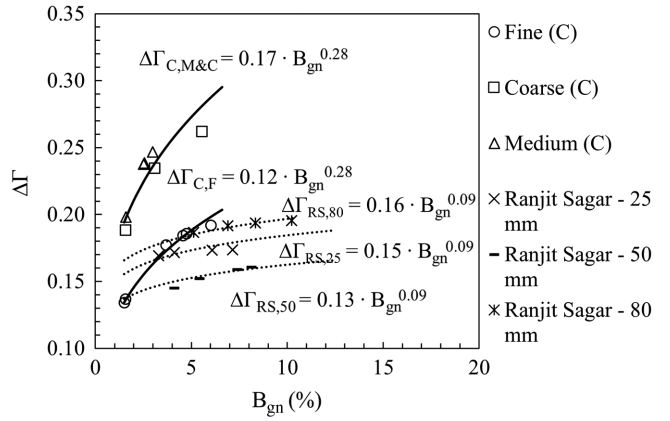


Fig. 27. Intercept Γ of the CSLs versus B_{gn} for saturated specimens of Coreno and Ranjit Sagar rockfills. (Data from Varadarajan et al. 2003.)

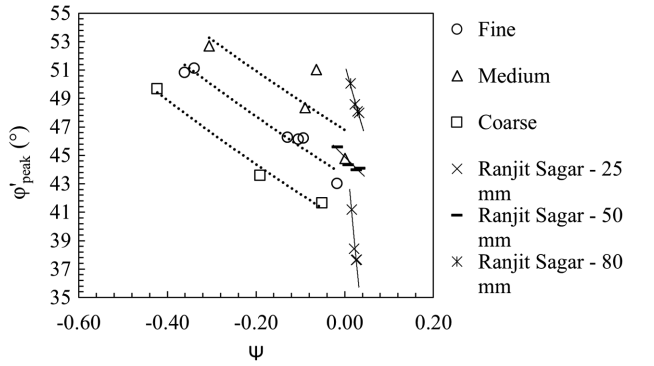


Fig. 28. Peak friction angle versus state parameter ψ for Coreno and Ranjit Sagar rockfills. (Data from Varadarajan et al. 2003.)

Despite the limited stress range tested in the experimental work ($p'_{max} < 1$ MPa), some conclusions can be drawn:

- The test results confirmed that particle breakage is a function of particle size and soil grading, mineralogy, degree of saturation, and stress path;
- The deviatoric component of the loading path is the major factor responsible for particle breakage;

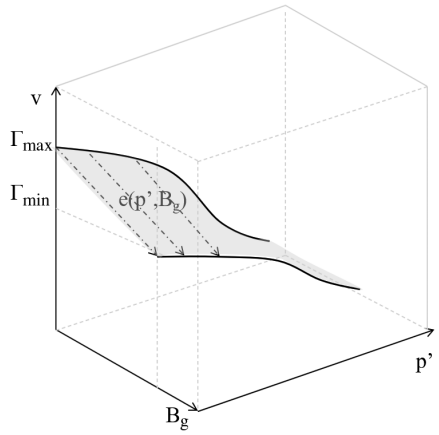


Fig. 29. Critical state surface in the v–p'–B_g space.

- A simplified elliptical iso-B_g locus can be identified in the deviatoric plane, and its shape is ruled by a single parameter—the ratio between the two axes of the ellipse, a:b (aspect ratio);
- The aspect ratio can be reasonably assumed as intrinsic to the type of soil considered;
- The evolution of the grain breakage locus on loading is mainly ruled by the initial coefficient of uniformity, which seems capable of effectively describing the micromechanical interparticle forces;
- The experimental data confirmed the existence of a critical state surface in the v–p'–B_g space (Fig. 29) that explains the existence of a multiplicity of CSLs in the compression plane correlated with the current grading; and
- Within the investigated stress range, the downward shift of the CSL in the compression plane v–$\ln(p')$ predicted by Eq. (7) ($\Delta\Gamma$) during shearing seems to mainly be ruled by particle breakage.

Appendix. A Simplified Micromechanical Analysis

It is well known that the change in crushable soil grading can be explained by micromechanical analysis evaluating the average contact stress at grain-to-grain contacts. To this end, the simplified procedure suggested by Flora et al. (2007) was applied.

The total number of vertical contacts N on a face of a generic horizontal plane of a specimen is defined as

$$N = N_{p,t} \cdot N_{c,p} \tag{9}$$

where $N_{c,p}$ = number of contacts in the vertical direction on one of the two sides of the considered plane; and $N_{p,t}$ = total number of particles calculated using the following relation:

$$N_{p,t} = 1/(\gamma_s \cdot \beta) \cdot \Sigma(P_i/d_{m,i}^3) \tag{10}$$

where γ_s = unit weight of solids; β = coefficient of proportionality between the volume of a single particle and its equivalent diameter (assumed equal to 0.3); $d_{m,i}$ = average diameter; and P_i = weight of the ith granulometric fraction that constitutes the specimen, respectively.

The term $N_{c,p}$ in Eq. (9) is a function of specimen porosity n:

$$N_{c,p} = 1/(2 \cdot n) \tag{11}$$

The average contact force F_i in the vertical direction acting on a single grain is, therefore, determined by the following relation:

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$$F_{i,v} = (\pi \cdot D^2 \cdot \sigma_v)/(4 \cdot N) \quad (12)$$

where D = diameter of the specimen; and σ_v = current vertical stress.

The total number of vertical contacts N calculated by Eq. (9) is about equal to $1.4 \cdot 10^6$ for the fine grain size distribution and $5.7 \cdot 10^5$ and $1.1 \cdot 10^4$ for the medium and coarse gradations, respectively. So, as expected, the fine gradation, which has a lower $d_{\max}$, also has a higher number of intergranular contacts.

Once the average contact force $F_{i,v}$ has been calculated using Eq. (12), the average contact stress on a single grain in the vertical direction $\sigma_{c,d}$ is expressed as

$$\sigma_{c,d} = F_{i,v}^{1/3}/[\pi \cdot K^{2/3} \cdot (0.5 \cdot d_{50})^{2/3}] \quad (13)$$

where $K = 3(1-\nu^2)/(4E)$; and ν and E = Poisson's ratio and Young's modulus of the rock, respectively. In this case, the values $\nu = 0.30$ and $E = 800$ MPa were used.

Data Availability Statement

The authors confirm that all data used during the study appear in the published article.

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